



GRID SURVIVAL

Multiplayer Game Development Project

Project Proposal

Level 2 Students

The ICT University, ICT Faculty

May 2026

gridsurvival.duckdns.org

1 Project Overview

1.1 Origin and Inspiration

The concept of Grid Survival emerged during our Game Development course at ICT University, where students were tasked with creating a fully functional, playable game with multiplayer support. Our team drew inspiration from the competitive minigame format popularized by Mario Party and similar party games, which combine simple controls, quick match duration (2–5 minutes), and intense competition.

We identified a gap in accessible, LAN-based competitive games suitable for classroom and campus settings, and designed Grid Survival to fill that niche: a game that is quick to learn, fun to play, and technically sophisticated enough to serve as a portfolio piece demonstrating modern game development practices.

1.2 Game Concept

Title: Grid Survival

Genre: Multiplayer Party Game

Platform: PC (Windows 10/11)

Player Count: 3–4 players per match

Match Duration: 5–15 minutes

Core Mechanics:

- Navigate a disappearing tile grid while avoiding environmental hazards
- Collect power-up orbs that grant temporary abilities
- Outlast opponents through skill, timing, and strategy
- Compete in Campaign, LAN, and Online modes

2 Project Requirements

2.1 Server Infrastructure (for Online Mode)

To support worldwide online play, the following server specification is required:

Specification	Requirement
Compute	2–4 vCPU (2.0+ GHz)
RAM	8–16 GB
Bandwidth	50–200 Mbps
Storage	SSD 50 GB (logs, assignment cache)
Concurrent Matches	25–125 (3–4 players each)
Operating System	Linux (Ubuntu 20.04+)
Network Protocol	UDP (port 5555)

Table 1: Server Specifications for Small-Scale Online Play (100–500 concurrent players)

3 Game Modes

Grid Survival supports four distinct gameplay modes, each optimized for different player counts and network environments:

3.1 Campaign Mode (Single-Player vs. AI)

Campaign Mode allows a single player to compete against AI-controlled opponents in a series of matches. This mode is ideal for:

- Learning the game mechanics without pressure
- Practicing character strategies
- Unlocking secret characters and achievements
- Solo entertainment

Features:

- 3–4 AI opponents with adjustable difficulty (Easy, Normal, Hard)
- Progression system with unlockable rewards
- Best of 3, 5, or 7 match format
- Local match history and statistics

3.2 Local Multiplayer

Local Multiplayer enables 2–4 players to compete on the same machine using shared keyboards, mice, or gamepads. Ideal for:

- Classroom tournaments
- Campus game hall competitions
- Couch gaming sessions
- Party events

Features:

- Split-screen or windowed display modes
- Full character and power-up availability
- Remappable controls for each player
- Real-time scoreboard and HUD
- Best-of-series match support

3.3 LAN Multiplayer

LAN Multiplayer connects multiple machines over a Local Area Network (campus network, office network, or home router). Ideal for:

- Campus tournaments and events
- LAN parties
- Classroom competitions
- Small-scale social gaming

Technical Details:

- Host discovers clients automatically via UDP broadcast
- No internet connection required
- Latency: Typically 1–50 ms on a stable LAN
- Supports up to 4 players per match (expandable with multiple match instances)
- Persistent lobby chat and team management

3.4 Online Multiplayer

Online Multiplayer connects players over the internet via our dedicated VPS infrastructure. Ideal for:

- Global player base
- Ranked competitive matches
- Asynchronous matchmaking (queue system)
- Persistent player accounts and statistics

Technical Details:

- Authoritative server architecture (cheat-proof)
- Server-side snapshot replication for smooth client-side interpolation
- Automatic match discovery and lobby creation
- Latency tolerance: Playable up to 150 ms ping
- Player account system with rating (RR), win/loss tracking, and leaderboards
- Support for 100–500 concurrent players (scales with server resources)

4 System Requirements

4.1 Player Machine Requirements

Players require different system specifications depending on desired performance and visual quality:

4.1.1 Minimum Specifications (30 FPS, Low Quality)

To run Grid Survival at playable frame rates with minimal graphics fidelity:

Component	Requirement
Processor	Intel Core i3 (6th Gen) or AMD Ryzen 3 1200
RAM	4 GB
GPU	Intel UHD Graphics 530 or equivalent
Storage	2 GB SSD
Display	1280×720 resolution, 60 Hz
Network	5 Mbps (LAN/Online)
Operating System	Windows 10 (Build 19041+), Windows 11, Ubuntu 20.04+

Table 2: Minimum System Specifications

4.2 On-Campus Installation

For local deployment in the school game hall:

- **Computer Specification:** Two Desktop/Laptop with minimum 4 CPUs, 4 GB RAM, SSD 2 GB
- **Display Setup:** Two monitors for multiplayer viewing
- **Input Devices:** Two keyboards (shared setup)
- **Input Devices:** Two mice (shared setup)
- **Network:** Stable LAN connection (1 Gbps recommended)
- **Location:** School game hall or computer lab

5 Project Budget and Infrastructure Costs

This section details the financial and technical investments required to deploy and operate Grid Survival online, including AI tooling for development and VPS infrastructure for operations.

5.1 VPS Infrastructure for Online Multiplayer

Grid Survival Online mode requires a dedicated Virtual Private Server to host authoritative game servers, matchmaking lobby, and player account database. Below is the detailed specification and cost breakdown:

5.1.1 VPS Specifications and Rental Agreement

Specification	Configuration
Provider	CONTABO (contabo.com)
Instance Type	VPS M (Virtual Private Server)
vCPU Allocation	4 vCPU @ 3.0+ GHz
RAM	8 GB DDR4
Storage	160 GB SSD (NVMe)
Bandwidth Allocation	Unlimited
Region	Frankfurt, Germany (Europe)
Operating System	Ubuntu 22.04 LTS
Public IPv4 Addresses	1 static IP & 1 IPv6 /64
Uptime Guarantee	99.9% SLA

Table 3: VPS Configuration and Technical Specifications (CONTABO VPS M)

5.1.2 VPS Rental Costs

Billing Period	Monthly Cost	Annual Cost
Monthly (\$15/month)	USD 15.00	USD 180.00
Annual pre-paid	–	USD 96.00

Table 4: VPS Rental Pricing (CONTABO VPS M — Frankfurt Data Center)

Current Project Agreement:

- **Billing Period:** Monthly at USD 9.00/month
- **Duration (Project Phase):** 3 months (March–May 2026)
- **Project Phase Total Cost:** USD 27.00
- **Planned Extended Duration:** 12 months (Beyond May 2026) at USD 96.00/year (annual pre-paid rate)
- **Extended Phase Total Cost:** USD 96.00

Item	Cost
Domain Registration (gridsurvival.duckdns.org)	Free (DuckDNS DDNS service)
SSL/TLS Certificate	Free (Let's Encrypt automation)
Database Backups (automated)	Included with VPS
Bandwidth Overage	Free (CONTABO provides unlimited bandwidth)
Monitoring & Alerting Tools	Free (CONTABO API + cron scripts)

Table 5: Additional VPS Operational Costs

Category	Project Phase (3 mo)	Year 1 (12 mo)
AI Tooling (3 developers × USD 10/month)	USD 90.00	USD 360.00
VPS Infrastructure Rental (CONTABO)	USD 27.00	USD 96.00 (annual pre-paid)
VPS Operational Overhead	Free	Free (unlimited bandwidth included)
Total Estimated Cost	USD 117.00	USD 456.00

Table 6: Project Budget Summary (AI Tooling + CONTABO VPS M)

5.1.3 Additional VPS Operational Costs

5.2 Total Project Budget Summary

Budget Notes:

- All costs are in USD; conversions to other currencies may apply depending on funding source
- Team member labor (development time) is not included; this section covers infrastructure and operational hosting costs only
- AI tooling estimate assumes 3 developers using subscriptions at USD 10/month
- VPS costs are sustainable for small-to-medium player bases (100–500 concurrent players)
- Scaling to larger player counts (1000+ concurrent) will require infrastructure upgrade to an enterprise-tier server (estimated USD 50–100/month)
- All financial transactions are documented and receipts are retained for institutional audit purposes

5.3 Total Expenditures Already Spent

As of May 2026, the project has already incurred the following direct costs:

Expense Item	Amount Already Spent
AI Tooling (3 developers × USD 10/month × 3 months)	USD 90.00
VPS Infrastructure Rental (CONTABO, 3 months)	USD 27.00
Total Expenditures Already Spent	USD 117.00

Table 7: Total Project Expenditures Already Spent (As of May 2026)

6 Team Composition and Roles

6.1 Development Team

Below is a detailed breakdown of each team member’s role and specific contributions to Grid Survival, including their contact details.

Kamdeu Yamdjeuson – Lead Engine & Network Developer

Kamdeu designed the core game loop and implemented the entire UDP-based networking layer. He built the authoritative server architecture, client-side prediction, and snapshot interpolation that allows smooth multiplayer even on imperfect connections. He also integrated the matchmaking lobby system and handles server deployment scripts.

Contact: Matricule [ICTU20241386] — Phone [676093910]

Ayukeyong Dohbila – UI Designer & Front-End Developer

Ayukeyong created all user interfaces from scratch: title screen, main menu, character selection, player cards, in-game HUD, and post-match results. He ensured the UI is responsive and visually consistent, and he implemented custom animations for buttons, transitions, and feedback effects.

Contact: Matricule [ICTU20234131] — Phone [653591460]

Lum Azinui Emerald – Sound Designer & Asset Builder

Lum composed the original background music and recorded all sound effects (footsteps, power-up pickups, tile falls, victory/loss jingles). She also sourced and optimised free-to-use assets for environmental tiles, hazards, and UI icons, and integrated the audio mixer with volume control and dynamic attenuation.

Contact: Matricule [ICTU20241334] — Phone [681065536]

Neba Ehud – Gameplay UI Developer

Neba focused on the in-game user interface elements: player health bars, power-up cooldown timers, scoreboards, and event notifications. He wrote the logic for displaying tile disappearance warnings and hazard indicators, and worked closely with Ayukeyong to ensure seamless UI integration with the game engine.

Contact: Matricule [ICTU20241812] — Phone [673674345]

Bissekie Ndongo – Character Artist & Sprite Animator

Bissekie designed all eight playable characters (Caveman, Giant Goblin, Ninja, Viking Leader, Chief Goblin, Female Goblin, and two secret characters). He created sprite sheets for idle, running, jumping, and dying animations, as well as particle effects for power-ups and explosions. His art style gives Grid Survival its distinctive, colourful look.

Contact: Matricule [ICTU20241537] — Phone [659253395]

Tabi Paul – Gameplay Mechanics Developer

Tabi implemented the core rules: tile disappearance system, hazard spawning, collision detection, power-up logic, and win/loss conditions. He balanced the game's difficulty across three modes (Campaign, LAN, Online) and coded the AI opponents for single-player campaign, ensuring they provide a fair but challenging experience.

Contact: Matricule [ICTU20241934] — Phone [650716480]

Chijioke Emmanuel – Assistant Audio Producer

Chijioke assisted Lum with audio production: cleaning recordings, looping background tracks, and implementing positional audio for hazards. He also managed the audio file versioning and ensured all sound assets met the game's performance targets (low memory footprint, fast loading).

Contact: Matricule [ICTU20141760] — Phone [Phone]

Abongong Patience – Community & Outreach Coordinator

Price manages the project's social media presence, Facebook Community (<https://www.facebook.com/share/g/1GT6LzbqPa/>), Discord server, and the official website (<https://gridsurvival.duckdns.org>). She collects player feedback, organises playtesting sessions, and will handle the public release announcement. She also writes documentation and user guides.

Contact: Matricule [ICTU20241375] — Phone [673046663]

Kwete Rayan – Game Tester & Error Finder

Rayan systematically tested gameplay mechanics and identified critical bugs to ensure a polished and stable player experience.

Contact: Matricule [ICTU20241377] — Phone [688435162]

7 Development Timeline

7.1 Current Progress (2 Months)

Development Start Date: March 2026

Current Date: May 2026

Duration Completed: 2 months

8 Game Features

- **Game Modes:** Campaign (AI), LAN Multiplayer, Online Multiplayer
- **Characters:** Caveman, Giant Goblin, Ninja, Viking Leader, Chief Goblin, Female Goblin (plus two unlockable secret characters)
- **Power-Ups:** Speed boost, shield, void walk, freeze, life extension
- **Hazards:** Falling tiles, water hazards, enemy Pac-Man-style AI roaming the grid
- **Customization:** Remappable controls (keyboard/gamepad), difficulty levels
- **Visual Polish:** Sprite animations, particle effects, smooth transitions

9 Conclusion

Grid Survival represents a successful application of modern game development principles, including networking architecture, real-time physics, multiplayer synchronization, and user experience design. The two-month development cycle demonstrates rapid iteration and team collaboration, while the technical depth (custom UDP networking, authoritative server, client-side smoothing) positions this project as a compelling portfolio piece.

With the requested server infrastructure and on-campus installation, Grid Survival will transition from a campus project to a playable, distributable product capable of supporting a growing player base.

9.1 Supervisor

Academic Supervisor: Engr Daniel Moune

Affiliation: ICT University, Faculty of Computing

Role: Technical guidance and infrastructure advice.

Contact: Email [moune.daniel@ictuniversity.edu]

Project Status: *Deployed — Beta Playable*

Website: gridsurvival.duckdns.org